

Matic Šuc

Software Engineer • AI & Backend

About

Developer with hands-on experience in computer vision, AI/ML engineering, simulation systems, and full-stack development. I've worked across research, product, and engineering contexts – from model training and data pipelines to APIs and on-site hardware setup. I like building things properly, end-to-end. Currently deep in AI experimentation – building things, testing ideas, and seeing what sticks.

Experience

AI Driving Metaverse Group, 2025 - 2026

Simulation Software Developer

- Developed driving scenarios in **SCANer Studio** with **Python** and maintained a **Flask API** connecting simulator services, handling data output, format standardization (CSV/JSON), and pipeline integration for downstream reporting.
- Set up and configured simulators on-site – hardware, OS, drivers, fully custom software stack, internal networking
- Maintained production simulator and managed software deployments across remote instances
- Delivered on-site customer training and remote technical support

TK Analytics, 2023 - 2024

Python AI Engineer

- Built **Python** pipelines for high-content image processing, profiled large-scale HCS datasets with **DeepProfiler** on **AWS S3**
- Researched, forked, and customized open-source computer vision models for internal analytical use
- Developed and maintained the company's **Python SDK** for model integration, data handling, and **AWS** inference optimization

Laboratory for Artificial Perception Systems, FRI, 2022 - 2023

Research Assistant

- Refactored an existing **CNN** from image-level classification to pixel-level segmentation in **PyTorch**, adding upsampling, SE blocks and a custom output layer
- Trained and evaluated the model across multiple datasets on FRI's GPU cluster (**CUDA/SLURM**)
- Benchmarked results across configurations and co-authored a paper accepted to **Construction and Building Materials**

Kolektor Vision, 2021 - 2022

Computer Vision Software Engineer

- Applied traditional computer vision methods to enhance production line quality control and automation
- Delivered end-to-end custom computer vision solutions for industrial production lines – from requirements gathering to deployment
- Extended the company's SDK to implement client-specific inspection, dimensional measurement, and object detection features

Projects

Journal

Journaling app with AI personas that reflect on your entries, built full-stack with **Next.js**, **Supabase**, and **Ollama**. Started as a personal experiment – now a daily-use product I keep improving.
journal-reflect.vercel.app • github.com/maticsuc/journal

Srečko Kosovel RAG

Scraped ~460 Kosovel's works into a **pgvector PostgreSQL** database with hybrid semantic and full-text search, powered by **LangChain**, **Ollama** for embedding model and **OpenRouter** for LLM inference. Built a conversational interface and chat UI for exploring his poetry in natural language, deployed on **Hetzner**.
178.104.200.66/srecko • github.com/maticsuc/srecko

Full CV

maticsuc.github.io/cv

Contact

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github.com/maticsuc

Education

Faculty of Computer and Information Science (FRI), University of Ljubljana

2022 - 2025

Master of Computer and Information Science

2018 - 2022

Bachelor of Computer and Information Science

Publications

Automated detection and segmentation of cracks in concrete surfaces using joined segmentation and classification deep neural network, Construction and Building Materials, 2023

Skills

AI/ML

PyTorch • Keras • TensorFlow • OpenCV • NumPy • Pandas • LangChain • Ollama • Hugging Face • Claude Code • OpenCode • OpenClaw

Backend & APIs

Python • Flask • FastAPI • Django • Quarkus • PostgreSQL • Supabase • RabbitMQ • Redis

Frontend

TypeScript • React • Next.js

Infrastructure & DevOps

Git • Linux • Bash/Shell • Docker • Nginx • Caddy • Tailscale • Vercel • Hetzner • AWS • Azure

Hardware & IoT

ESP32 • RaspberryPi • Arduino

Other Languages

Java • C++ • Powershell • SQL

Languages

Slovenian • Native
English • Fluent
Italian • Intermediate

Mini Projects

BP Tracker

Personal blood pressure tracker built with **Next.js**, **Prisma**, **PostgreSQL**, and **Supabase Auth**. Deployed on **Vercel** – I actually use it daily to monitor my own readings.

Status Board

ESP32 with **I2C display**, **temp** and **touch sensors** acting as my desktop status board, running 24/7.